Grant Proposal Assessment Part A: Preliminary Phylogeny

Student Name – Student number

Trait Evolution Across Cockatoo Phylogeny:

The maximum size that a cockatoo might reach varies drastically across the taxon. Within the Cacatuidae family, there are two large subfamilies: Calyptorhynchinae and Cacatuinae. The Calyptorhynchinae subfamily includes the genera *Calyptorhynchinae* and *Zanda*, which are closely related (Figure 1). The maximum size trait in this subfamily has a relatively narrow range (17cm) compared to the Cacatuinae subfamily (38cm) (Table 1). The Calyptorhynchinae subfamily diverged first based on this phylogeny and has maintained a relatively stable maximum size across species, averaging 58.25cm. In contrast, a large variation in maximum size is seen across the Cacatuinae subfamily. This subfamily contains three genera, including the largest species (*Probosciger aterrimus*, 68cm) and the smallest species (*Cacatua ducorpsii*, 30cm). The *Probosciger* genus branches off first within this subfamily, suggesting that the common ancestor of the Calyptorhynchinae subfamily and *Probosciger* genus likely had a maximum size between 58.25 and 68cm.

Following this, the *Eolophus* and *Cacatua* branch out in the Cacatuinae subfamily. These birds are smaller than those in the *Probosciger* genus, averaging 35cm and 40.79cm respectively. Within the *Cacatua* genus, there are two subgenera. It is seen on this phylogeny that the *Licmetis* subgenus has a smaller maximum body size compared to the *Cacatua* subgenus.

Overall, the phylogeny suggests that the common ancestor of these genera likely had a body size similar to that of the Calyptorhynchinae subfamily, which is the first to branch off and exhibits a relatively stable maximum size. Subsequent diversification led to the development of both larger and smaller species, with more extant species exhibiting smaller maximum body sizes.

Figure 1. A phylogenetic tree created in RStudio (version 2024.04.1) of the family Cacatuidae, including the subfamilies Calyptorhynchinae and Cacatuinae, as well as an outgroup subfamily species, Nymphicus hollandicus.

Table 1. Maximum size data across all species, including averages of genera and subfamilies.

Scientific Name	Maximum Size (cm)	Genus Mean (cm)	Subfamily Mean (cm)	Subfamily Range (cm)	Overall Mean (cm)	Overall Range (cm)
Nymphicus hollandicus	32	32	32	32	47.69	38
Calyptorhynchus banksii	60	57.6	58.25	17		
Calyptorhynchus banksii graptogyne	60					
Calyptorhynchus banksii naso	60					
Calyptorhynchus banksii samueli	60					
Calyptorhynchus lathami	48					
Zanda baudinii	57	59.33				
Zanda funerea	65					
Zanda latirostris	56					
Probosciger aterrimus	68	68	43.65	38		
Probosciger aterrimus goliath	68					
Eolophus roseicapilla	35	35				
Cacatua alba	46	40.79				
Cacatua ducorpsii	30					
Cacatua galerita	55					
Cacatua galerita eleonora	44					
Cacatua goffiniana	31					
Cacatua haematuropygia	31					
Cacatua moluccensis	50					
Cacatua ophthalmica	50					
Cacatua pastinator butleri	45					
Cacatua sanguinea	40					
Cacatua sulphurea	35					
Cacatua sulphurea sulphurea	35					
Cacatua tenuirostris	40					
Lophochroa leadbeateri	39					

Table 2. Accession ID numbers for all species used in the phylogenetic tree.

Accession ID	Name
694881298	Cacatua_sulphurea_sulphurea
694881296	Cacatua_galerita
694881112	Cacatua_sanguinea
694881110	Cacatua_haematuropygia
694881108	Cacatua_galerita_eleonora
694880988	Probosciger_aterrimus_goliath
694880986	Calyptorhynchus_banksii
694880982	Zanda_funerea
694880978	Lophochroa_leadbeateri
694880974	Cacatua_tenuirostris
694880972	Cacatua_ophthalmica
33340959	Nymphicus_hollandicus
33340955	Eolophus_roseicapilla
33340949	Probosciger_aterrimus
330686630	Cacatua_pastinator_butleri
330686618	Calyptorhynchus_lathami
330686616	Calyptorhynchus_banksii_samueli
330686614	Calyptorhynchus_banksii_naso
330686608	Zanda_baudinii
330686604	Zanda_latirostris
1938493742	Cacatua_ducorpsii
121308521	Cacatua_moluccensis
121308513	Cacatua_sulphurea
121308511	Cacatua_goffiniana
121308507	Cacatua_alba
1036029355	Calyptorhynchus_banksii_graptogyne

Table 3. Code used in RStudio to create the phylogeny.

```r
#Assignment

RootPath <- "C:/Users/tegan/OneDrive/Desktop/BIOL365_Pracs_2024"
setwd(RootPath)

install.packages("dplyr")
install.packages("magrittr")
install.packages("rentrez")
install.packages("ape")

library(dplyr)
library(magrittr)
library(rentrez)
library(ape)
```

```
Cockatoo_Search <- rentrez::entrez_search(db = "nucleotide",
                                          term = "(cytb[Gene Name]) AND
(Cacatuidae[Organism])",
                                          retmax = 200)
Cockatoo_Search
#returned 116 hits

Cockatoo_Cytb <- ape::read.GenBank(access.nb = Cockatoo_Search$ids)
class(Cockatoo_Cytb)
#this is a DNAbin class

Cockatoo_Cytb
base::saveRDS(Cockatoo_Cytb,
              file = "Cockatoo_Cytb.rds")
ape::write.FASTA(Cockatoo_Cytb,
                 file = "Cockatoo_Cytb.fasta")

install.packages("BeeBDC")
library(BeeBDC)
library(magrittr)

install.packages("taxadb")
library(taxadb)
speciesCountTable <- BeeBDC::taxadbToBeeBDC(name = "Cacatuidae",
                                            rank = "Family",
                                            provider = "gbif") %>%
  dplyr::filter(taxonomic_status == "accepted") %>%
  dplyr::group_by(genus) %>%
  dplyr::count()

class(speciesCountTable)
speciesCountTable
#this gives the list of the genus' in Cacauidae. Thats all for week 2 stuff.

install.packages("tibble")
install.packages("tidyr")
install.packages("stringr")
install.packages("stringi")

library(tibble)
library(tidyr)
library(stringr)
library(stringi)

Cockatoo_Cytb_data <- base::readRDS("Cockatoo_Cytb.rds")

seqLengths <- lapply(Cockatoo_Cytb_data, length) %>%
  as.numeric()
```

```
print(seqLengths)

genBankSummaryTibble <- tibble::tibble(
  accessionID = attributes(Cockatoo_Cytb_data)$names,
  species = attributes(Cockatoo_Cytb_data)$species,
  seqLength = seqLengths,
  description = attributes(Cockatoo_Cytb_data)$description,
)

genBankSummaryTibble

theChosenSeqs <- genBankSummaryTibble %>%
  dplyr::filter(seqLength < 1500) %>%
  dplyr::filter(!stringr::str_detect(description, "control region")) %>%
  dplyr::group_by(species) %>%
  dplyr::arrange(species, dplyr::desc(seqLength))

theChosenSeqs

theChosenSeqs <- theChosenSeqs %>%
  dplyr::filter(dplyr::row_number()==1)

names(Cockatoo_Cytb_data)
Cockatoo_Cytb_data[1]

CockatooSpecies <- Cockatoo_Cytb_data[theChosenSeqs$accessionID]

attributes(CockatooSpecies)$names <- stringr::str_c(theChosenSeqs$species,
                                                     theChosenSeqs$accessionI
D,
                                                     sep = "_")

CockatooSpecies

library(msaR)
library(BiocManager)
library(Biostrings)
library(msa)
library(dplyr)

CockatooSpecies_fasta <- msaR::as.fasta(CockatooSpecies)

lengthCorrectionTable <- tibble::tibble(sequence = CockatooSpecies_fasta)
%>%
  tidyr::separate_longer_delim(cols = "sequence",
                               delim = ">") %>%
  dplyr::filter(!dplyr::row_number() == 1) %>%
```

```
  dplyr::as_tibble() %>%
  dplyr::mutate(sequence = sequence %>% stringr::str_remove("\\n$")) %>%
  tidyr::separate_wider_delim(sequence, delim = "\n",
                               names = c("name", "sequence")) %>%
  dplyr::mutate(lengthenedSequence = stringr::str_pad(sequence, pad = "-",
side = "right",
                                                       width =
max(stringr::str_count(sequence))),
                length2 = stringr::str_count(lengthenedSequence))

BiocManager::install("Biostrings")
BiocManager::install("msa")

library(Biostrings)
library(msa)

equalisedFasta <- stringr::str_c(">", lengthCorrectionTable$name,"\n",
                                 lengthCorrectionTable$lengthenedSequence,
                                 collapse = "\n")
write(equalisedFasta, "equalisedFasta.fasta")
equalised_BioString <- Biostrings::readDNAStringSet("equalisedFasta.fasta")


msaR::msaR(equalised_BioString, menu = T, overviewbox = T, labelNameLength =
200)

alignedCockatoos <- msa::msa(equalised_BioString)

msaR::msaR(alignedCockatoos@unmasked, menu = T, overviewbox = T,
labelNameLength = 200)

alignedDNAbin <- ape::as.DNAbin(alignedCockatoos@unmasked)
distanceCockatoos <- ape::dist.dna(alignedDNAbin)

distanceCockatoos

njCockatoos <- ape::nj(distanceCockatoos) %>%
  ape::root(outgroup = "Nymphicus_hollandicus_33340959")

njCockatoos

plot(njCockatoos)

ape::bionj(distanceCockatoos) %>%
  ape::root(outgroup = "Nymphicus_hollandicus_33340959") %>%
  plot()

library(phangorn)
```

```r
aligned_phyDat <- phangorn::phyDat(alignedDNAbin)
Cockatootest <- phangorn::modelTest(aligned_phyDat,
                                    model=c("JC", "F81", "K80", "HKY",
"SYM"),
                                    I = FALSE, G = FALSE)

#HKY fits best

fit_mt <- phangorn::pml_bb(Cockatootest,
                           control = phangorn::pml.control(trace = 0))

fit_mt


bs <- phangorn::bootstrap.pml(fit_mt,
                              bs=100, optNni = TRUE,
                              method = "ultrametric",
                              control = phangorn::pml.control(trace=0))

phangorn::plotBS(
  tree = bs %>%
    phangorn::maxCladeCred() %>%
    ape::root(outgroup = "Nymphicus hollandicus"),
  BStrees = bs,
  p = 10, type="p", digits=2, main="Ultrafast Bootstrap")


ape::write.FASTA(alignedDNAbin,
                 "alignedDNAbin.fasta")

names(alignedDNAbin) <- names(alignedDNAbin) %>%
  stringr::str_remove_all("HQ[0-9]|[0-9]") %>%
  stringr::str_replace_all("_", " ") %>%
  stringr::str_squish()

names(alignedDNAbin)

CockatooML <- alignedDNAbin %>%
  phangorn::phyDat() %>%
  phangorn::pml_bb(model = "HKY",
                   control = phangorn::pml.control(trace=0)) %>%
  phangorn::bootstrap.pml(bs = 100, optNni = TRUE,
                          method = "ultrametric",
                          control = phangorn::pml.control(trace = 0))

(treeML <- phangorn::plotBS(
  tree = CockatooML %>%
```

```
    phangorn::maxCladeCred() %>%
    ape::root(outgroup = "Nymphicus hollandicus"),
  BStrees = CockatooML,
  p=10, type="p", digits = 2, main="Ultrafast bootstrap")
)

cockatoosize <- readxl::read_excel("Data.xlsx",
                                   sheet = "Sheet1",
                                   guess_max = 7000)

dnaNames <- names(alignedDNAbin) %>%
  tibble::tibble(scientificName = .)

compareNameTable <- dplyr::left_join(dnaNames, cockatoosize, by =
"scientificName")
compareNameTable
library(phytools)

treeML_dropped <- ape::drop.tip(treeML,
                                tip = "Cacatua sp. CCJ-")

cockatoosize <- cockatoosize %>%
  dplyr::filter(scientificName %in% treeML_dropped$tip.label)

treeML_fixed <- treeML_dropped

treeML_fixed$edge.length <- treeML_fixed$edge.length %>%
  dplyr::if_else(. < 0.001, 0.001, .)

treeML_fixed <- treeML_fixed %>%
  ape::multi2di(random = TRUE, tol = 0.001)

par(mfrow = c(1,2))

plot(treeML_dropped, main = "original plot", sub = "node #s shown",
     col.sub = "darkred", cex = 0.5)
ape::nodelabels(treeML_dropped$node.label, bg = "white", frame = "circle",
cex = 0.5)
plot(treeML_fixed, main = "New plot", sub = "now with tiny branches added",
     col.sub = "darkred", cex = 0.5)

par(mfrow = c(1, 1))

pic_size <- ape::pic(cockatoosize$Size,
                     treeML_fixed)

sizevector <- setNames(cockatoosize$Size, cockatoosize$scientificName)
```

```r
ancSize <- phytools::fastAnc(treeML_fixed, sizevector,
                             vars = TRUE, CI = TRUE)

phytools::plotTree(treeML_fixed, ftype = "i", fsize = 0.5, lwd = 1)
phytools::labelnodes(text = ancSize$ace %>% round(0) %>% as.character(),
                     node = names(ancSize$ace) %>% as.numeric(),
                     interactive = FALSE,
                     shape = "circle",
                     cex = 0.5)

plotMap <- phytools::contMap(treeML_fixed, sizevector, plot = FALSE) %>%
  phytools::setMap(c("black", "purple", "white"))

plot(plotMap, leg.txt = "Maximum size (cm)", fsize = 0.7)
```

References:

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NSW Government Environment and Heritage (2009) *Red-tailed black-cockatoo (inland subspecies) (Calyptorhynchus banksii samueli) - vulnerable species listing*. NSW Government. Available at: https://www.environment.nsw.gov.au/topics/animals-and-plants/threatened-species/nsw-threatened-species-scientific-committee/determinations/final-determinations/2008-2010/red-tailed-black-cockatoo-inland-calyptorhynchus-banksii-samueli-vulnerable-species-listing (Accessed: 21 August 2024).

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